
CMSC472 Project Proposal

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1 Motivation

A computerized tomography (CT) scan is an imaging technique that uses X-rays to create cross-sectional images of the bones, blood vessels, and soft tissues of the body. CT scans are incredibly useful for diagnosing and monitoring conditions that affect these body parts. CT scans must be read by specialized doctors called radiologists. However, because these scans require human interpretation, there is the possibility for gross misinterpretation and error to be introduced. A study found that major interpretive discrepancies increased with longer shift times and a greater volume of scans to read. This ultimately is detrimental to not only the radiologists but more importantly the patients that they serve. We wanted to create a tool to help analyze CT scans in order to suggest a preliminary finding to the radiologist and therefore aid in the analysis of scans and diagnosing of patients.

2 Workshop Information

2.1 Workshop Challenge

Our project idea comes from the following CVPR 2026 Workshop on General CT Image Diagnosis

<https://fmv-cvpr26workshop.github.io/>.

<https://www.codabench.org/competitions/12650/>.

2.2 Data Source

We obtained our data sources from the workshop mentioned above. The datasets provided contain 10000+ CT imaging scans for pretraining and classification datasets for the downstream task. Additionally, the challenge provides links to additional training data which have been used to train baseline models. The following links provide the dataset source.

<https://huggingface.co/datasets/FLARE-MedFM/FLARE-Task4-CT-FM>.

<https://huggingface.co/datasets/kmin06/CVPR26-3DCTFMCompetition>.

<https://docs.google.com/spreadsheets/d/1xfHTQS6iJYqGwi3jZvWdFtFUqQ70eXKTgyQ8arvIk6A/>.

2.3 Baselines

The workshop provides the following baseline models and corresponding papers.

3DINO: <https://www.nature.com/articles/s41746-025-02035-w>
Code: <https://github.com/AICONSLab/3DINO>

SPECTRE: <https://arxiv.org/abs/2511.17209>
Code: <https://github.com/cclaess/SPECTRE>

Merlin: <https://www.nature.com/articles/s41586-026-10181-8>
Code: https://github.com/yws0322/Merlin_CVPR26CTFM

Curia: <https://arxiv.org/abs/2509.06830>
Code: <https://github.com/raidium-med/curia>

VoCo: <https://ieeexplore.ieee.org/document/11274411> Code:
https://github.com/kmin940/Large-Scale-Medical_CVPR26CTFM

3 Related Papers

Pai et. al(2024) developed a foundational model for cancer imaging biomarkers by pretraining an encoder on 11,467 CT lesions using self-supervised contrastive learning and then adapting it to downstream tasks, focusing mainly on malignancy prediction. Their work demonstrated that linear classification in limited data settings proved effective. Our project hopes to achieve similar results on 3D CT volumes as opposed to lesion patches and also target broader multi-organ classification.

Thakral et.al(2025) investigate the positive impact of pre-processing techniques on image enhancement, noise reduction, and feature extraction when applied to low-dose CT scans. While their specific methods differ from ours, their findings underscore the necessity of a robust preprocessing pipeline and its effects on downstream classification. This motivates our inclusion of Hounsfield Unit clipping and Isotropic Resampling prior to feature extraction.

4 Project Plan

Our proposal is to create a model to help detect anomalies within CT scans. CT scans are used to diagnose a multitude of conditions so we plan to try and create a model to detect anomalies in scans. Our achievable plan is to create a model to diagnose one specific condition like cancer. Our moon-shot plan is to create a model that can detect and diagnose different conditions and compete with the baselines on benchmarks.

4.1 Achievable Plan

Our primary objective is to create a Multi Organ Cancer Detection model. While existing models such as Merlin or 3DINO work great, they're both specific to a subset of possible scans. Merlin is specialized for abdominal scans, while 3DINO is limited to 10 specific organs. Our model will aim to generalize across a border anatomical range by using these foundational models provided in the workshop. The components of our plan are as follows:

- **Transfer Learning:** To do this, we can start with architecture and transfer learning using the 3DINO and SPECTRE models as a backbone for feature extraction. These models already have learned the 3D medical imaging jargon, so instead of starting from square one, we can instead fine tune the final layers specifically for the task of identifying malignant vs benign growth across multiple organ systems.
- **Data Integration:** Referencing Section 2.2, we will use readily available medical imaging data such as FLARE-TASK4-CT-FM or CVPR26-3DCTFM Competition datasets to provide our model with a diverse set of data. To ensure our model isn't biased towards any high frequency pathologies, we can implement an oversampling method for rarer cancer types found in our training data, or weighted cross-entropy loss.
- **Preprocessing Pipeline:** Since CT scans vary in resolution, we can develop a standardized preprocessing pipeline that can highlight softer tissues and also Isotropic Resampling to ensure 3D convolution kernels can process physical dimensions consistently.
- **Feasibility:** Focusing on a binary classification across multiple organs can ensure that the project is computationally feasible but providing more utility than a single organ baseline.

4.2 Moon-shot Plan

Our moon shot goal is to develop a Unified Multi Pathology Diagnostic System that can identify a wide variety of anomalies through different organ systems rather than focusing on just one condition. This would involve the following multi label classification, zero shot generalization, and Explainable AI with a report generation:

- **Multi Label Classification:** Moving from binary cancer detection to a model that is capable of identifying dozens of pathologies such as hemorrhages, fractures, lesions, etc. within a single scan would be ideal here.
- **Zero Shot Generalization:** Zero shot generalization would mean we leverage foundation models that were mentioned in Section 2.3 such as Merlin or 3DINO to make an ensemble model where we identify rare conditions that aren't explicitly represented in the fine tuning set.
- **Explainable AI:** With Explainable AI, rather than providing a simple probability score, the model would generate a saliency map of some sort where it highlights anomalous regions that may make it easier for health professionals to identify health vulnerabilities before onset.
- **Report Generation:** The report generation would entail natural language description of the findings which can mimic the output of a health professional and take the human out of the loop.

4.3 Evaluation Protocols and Metrics

To assess the performance of our Multi Organ Cancer Detection model, we can use a combination of statistical classification metrics and benchmarking protocols. Since medical datasets are usually highly imbalanced with significant more healthy scans than positive cases, we can prioritize metrics that account for this skewed distribution.

4.3.1 ROC-AUC

Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC) are metrics that are commonly used for evaluating medical related classification models, specifically for assessing the ability to diagnose conditions and diseases. The ROC curve compares the True Positive Rate (TPR) curve and the False Positive Rate (FPR) curves at multiple different thresholds to evaluate how well models are able to separate the different classes. AUC is a metric that pairs with ROC to define the probability the model will rank the positive classification over the negative. When considering medical conditions involving CT scans, it is generally considered better to prioritize sensitivity/recall over specificity. AUC will allow us to capture this sensitivity within our evaluation of our model.

4.3.2 F1 scores

F1 is a metric that evaluates classification models. It combines precision and recall into one metric, providing a tool that is especially useful for evaluating models on imbalanced datasets. To compute the F1 score, precision and recall must first be calculated. Precision examines the ratio of correct positives to all positive identifications. Recall, also known as sensitivity, compares the number of true positives to the number of true positives and false negatives. The F1 score is then computed by using the following equation:

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (1)$$

The higher the F1 score, the higher the precision and recall. If either metric dips, then the F1 score will decrease as well. Using F1 scores for analysis would allow us to see how well our model can positively identify cancer or any disease that is visible on the CT scans.

4.3.3 Balanced Accuracy

In addition to F1 Scores, Balanced Accuracy is another metric which can be used to evaluate models trained on imbalanced datasets. In order to calculate balanced accuracy, we first need to calculate

sensitivity/TPR and specificity/TNR and take the mean. Sensitivity is calculated by taking the ratio of true positive classifications over all positive samples. Specificity on the other hand, is the ratio between the true negative predictions over all the negative samples.

$$BalAcc = \frac{TPR + TNR}{2} \quad (2)$$

4.3.4 Technical Methodology and Pipeline

Section 3.1 outlines our goals using a preprocessing pipeline to process high dimensional 3D CT images and adapting foundation models for multi organ detection.

- **Data Preprocessing and Standardization:** CT data will be provided in DICOM or NIfTI formats which will require specialized handling. Accordingly, our pipeline will include Hounsfield Unit Clipping so that we can normalize the raw CT values to a range that's understandable for the model. Specifically, this will help for soft tissues that come up ambiguous for some vision models, and can also help to remove noise from bone and air pockets. Isotropic Resampling was also mentioned, and using the n4n-net (Isensee et al., 2021) framework as a guide, we can resample all the volumes to a consistent voxel spacing as referenced in the original paper. This will ensure that 3D kernels in our foundation models can maintain spatial consistency across various scan resolutions.
- **Fine Tuning Approach:** For finetuning, we can employ a Parameter-Efficient Fine-Tuning approach which allows us to essentially freeze weights and learn new tasks on top of the pre-existing model knowledge, an approach that's widely used in Machine Learning projects. In our case, we can freeze the encoder weights of the 3D Vision Transformer and insert LoRA layers to adapt the model to specific textures of cancerous lesions that it may not know prior. This way, we can implement a Multi-Head classification layer where each head corresponds to a specific organ system and allows for multi organ detection as per our achievable plan.

4.3.5 Running time

Running time measures how long the model takes to process and generate predictions. Unlike the other metrics, which evaluate the predictive performance of the model, running time measures its computational efficiency in inference. This is particularly relevant in the context of CT scans, which are large 3D volumes that can be very expensive to process, making efficiency an important factor for real-world application. For this specific workshop, running time will also measure the docker starting time.

5 Ethics and Safety

Deploying AI in a clinical setting comes with significant ethical responsibilities and overhead to make sure AI is used in a beneficial but safe way. Medical datasets often over represent certain demographics, which is important as our model will be trained using readily accessible data. We'll make sure to audit our model's performance across different pieces of metadata to ensure it doesn't have higher false-negative rates for specific populations due to a lack of data and ensure fairness without bias. Additionally, our tool is designed as a support system for decision making, not a total replacement for radiologists. By providing something like saliency maps, we ensure the radiologist remains the final authority, mitigating and risks for black box automation. All datasets from the workshop are also de-identified, and we'll maintain a strict adherence to data usage agreements. This will ensure patient re-identification is not possible through any markers in our data.

6 Citations

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